IMovies – Functional Specification Document

1. Introduction

- **Project title:** IMovies – Movie Ticket Booking Platform (MERN stack)
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2. Project Overview

Purpose. IMovies lets customers find a movie, pick exact seats for a show and pay online, while admins run the catalogue, theatres, show schedule and bookings from a separate panel.

Customer features

- Register, log in, edit profile and change password
- Browse, search and filter movies by genre, language, now showing / coming soon, with sorting
- Movie page with cast, rating, reviews and showtimes by city and date
- Interactive seat map (max 10 seats, live availability) with a 10-minute seat hold and countdown
- Simulated payment by card, UPI, net banking or wallet
- E-ticket with booking ID, printable; booking history; cancellation up to 2 hours before the show with simulated refund
- Star ratings and reviews; favourites and watchlist (email when a watchlisted movie gets a show)

Admin features (at `/admin/login`)

- Dashboard: revenue, tickets sold, top movies, last 7 days
- Manage movies, theatres and screens (custom seat layouts)
- Schedule shows with per-seat-type pricing and overlap protection
- View and cancel all bookings; activate, deactivate or delete users

3. Architecture

Frontend (React 19 + Vite). A single-page app in plain JavaScript/JSX using React Router v6. `App.jsx` defines the routes; `RequireAuth` guards customer pages (`role="user"`) and the admin area (`role="admin"`). Login state lives in `AuthContext`, API calls go through `api.js`, and shared UI sits in `components/` (Navbar, Footer, MovieCard, Poster, ui.jsx). The Vite dev server proxies `/api` to port 5000, so the browser sees one origin and the login cookie works without CORS setup.

Backend (Node.js + Express 4). `server.js` configures the app (helmet, CORS, JSON parser, cookie-parser, morgan, rate limiting on login/register) and mounts eight route groups under `/api`: auth, admin, movies, theatres, shows, bookings, reviews and favourites. Controllers hold the logic; middleware handles authentication, role checks, ObjectId validation and centralised errors. `utils/` holds seat logic, token generation, the simulated payment gateway, email sending and the seed script.

Database (MongoDB with Mongoose 8). Eight collections:

Collection	Key fields	Relations
User	name, email (unique), hashed password, phone, isActive	–
Admin	name, email (unique), hashed password	–
Movie	title, description, genre[], language, duration, certificate (U/UA/A/S), cast[], releaseDate, poster, trailerUrl, averageRating, ratingCount, isActive	status (upcoming / now showing) is derived from releaseDate
Theatre	name, city, address, screens[] (name + row layout: row, seats, type Regular/Premium/Recliner), isActive	embeds screens
Show	screenId, screenName, startTime, endTime, prices per seat type, bookedSeats[], isActive	movie, theatre
Booking	bookingCode, seats[] (seatId, type, price), totalAmount, status (pending/confirmed/cancelled/expired), expiresAt, payment (method, status, transactionId, paidAt)	user, show, movie, theatre
Review	rating 1–5, comment (max 1000 chars); one per user per movie	user, movie
Favourite	type (favourite / watchlist); unique per user, movie and type	user, movie

4. Setup Instructions

Prerequisites

- Node.js 18 or newer (check with `node -v`)
- MongoDB: either MongoDB Community Server installed locally, or a free MongoDB Atlas cluster whose connection string goes in `MONGO_URI`

Installation

1. Extract the project and open two terminals, one in `server/` and one in `client/`.
2. In `server/`, run `npm install`.
3. Open `server/.env` and set the variables below (change `JWT_SECRET` to a long random string).
4. Run `npm run seed` to create the admin account, sample movies, a theatre and shows (it only seeds an empty database).
5. In `client/`, run `npm install`.

Variable (server/.env)	Purpose
`PORT`	API port (default 5000)

Variable (server/.env)	Purpose
NODE_ENV	development or production (production makes cookies secure; sameSite=none)
MONGO_URI	MongoDB connection string
JWT_SECRET	Secret used to sign tokens
JWT_EXPIRES_IN	Token lifetime (default 7d)
CLIENT_URL	Allowed CORS origin (default http://localhost:5173)
SEAT_HOLD_MINUTES	Seat hold while payment is pending (default 10)
ADMIN_NAME, ADMIN_EMAIL, ADMIN_PASSWORD	Default admin created by the seed script
SMTP_HOST, SMTP_PORT, SMTP_USER, SMTP_PASS, EMAIL_FROM	Optional email settings; if SMTP_HOST is empty, emails print in the server console

Optional in client/: VITE_API_URL is needed only when the API is hosted on a different origin.

5. Folder Structure

Client (client/)

```
client/
  index.html, vite.config.js, package.json, .env.example
  public/favicon.svg
  src/
    main.jsx, App.jsx        entry point and routes
    api.js, hooks.js, utils.js
    index.css
    context/AuthContext.jsx  login state
    components/              Navbar, Footer, MovieCard, Poster, ui.jsx
    pages/                   Home, Movies, MovieDetail, SeatSelection, Checkout,
                             BookingDetail, MyBookings, Favourites, Profile, Login, Register
    pages/admin/             AdminLogin, AdminLayout, Dashboard, AdminMovies,
                             AdminTheatres, AdminShows, AdminUsers, AdminBookings
```

Server (server/)

```
server/
  index.js      entry point: loads .env, connects DB, starts server
  server.js     Express app, middleware and route mounting
  config/db.js  MongoDB connection
  models/       user, admin, movie, theatre, show, booking, review, favourite
  controllers/  auth, admin, movie, theatre, show, booking, review, favourite
  routes/       one router per controller
  middleware/   authMiddleware, validateObjectId, errorMiddleware
  utils/        seatUtils, generateToken, paymentGateway, sendEmail, seed, AppError, asyncHandler
```

6. Running the Application

Start the backend first, then the frontend, each in its own terminal.

Part	Directory	Command	Address
Backend (development)	server/	npm run dev (nodemon)	http://localhost:5000
Backend (plain)	server/	npm start	http://localhost:5000
Frontend (development)	client/	npm run dev	http://localhost:5173

The project uses Vite, so the frontend command is npm run dev rather than npm start. Check the API at /api/health. For production, run npm run build in client/ and serve client/dist from any static host.

Role	Login URL	Credentials
Customer	/register	any email and password you create
Admin	/admin/login	admin@imovies.com / Admin@123 (seeded demo account)

7. API Documentation

Base path /api. Every response is JSON with a success flag; errors return { "success": false, "message": "..." }. Access: **Public** = no login, **User** = customer login, **Admin** = admin login, **Any** = either.

Method	Endpoint	Access	Purpose / parameters
GET	/health	Public	API status check
POST	/auth/register	Public	Create account: name, email, password, phone
POST	/auth/login	Public	Log in: email, password
POST	/auth/logout	Public	Clear the login cookie
GET	/auth/me	Any	Current account
PUT	/auth/me	Any	Update profile / change password
POST	/admin/login	Public	Admin login: email, password
GET	/admin/dashboard	Admin	Revenue, tickets sold, top movies, last 7 days
GET	/admin/users, /admin/users/:id	Admin	List / view users
PUT, DELETE	/admin/users/:id	Admin	Activate or deactivate / delete a user
GET	/admin/bookings	Admin	All bookings
GET	/movies	Public	Query: search, genre, language, status, sort, page, limit
GET	/movies/:id	Public	Movie details
POST, PUT, DELETE	/movies, /movies/:id	Admin	Create / update / delete a movie

Method	Endpoint	Access	Purpose / parameters
GET	/theatres	Public	Query: city
GET	/theatres/:id	Public	Theatre with screens
POST, PUT, DELETE	/theatres, /theatres/:id	Admin	Manage theatres
POST, PUT, DELETE	/theatres/:id/screens[/:screenId]	Admin	Manage screens and seat layouts
GET	/shows	Public	Query: movie, theatre, city, date
GET	/shows/:id	Public	Show with seat map and availability
POST, PUT, DELETE	/shows, /shows/:id	Admin	Schedule / edit / remove shows
POST	/bookings	User	Hold seats (status pending): show, seats
POST	/bookings/:id/pay	User	Pay: method (card, upi, netbanking, wallet) and details
GET	/bookings/my	User	Own bookings
GET	/bookings/:id	Any	One booking (owner or admin)
PUT	/bookings/:id/cancel	Any	Cancel up to 2 hours before the show
GET	/reviews/movie/:movieId	Public	Reviews for a movie
POST	/reviews	User	Add review: movie, rating (1–5), comment
PUT	/reviews/:id	User	Edit own review
DELETE	/reviews/:id	Any	Delete (owner or admin)
GET	/favourites	User	Query: type (favourite / watchlist)
POST	/favourites	User	Body: movie, type
DELETE	/favourites/:movieId	User	Query: type

Example: login

```
POST /api/auth/login
{ "email": "user@example.com", "password": "secret123" }

200 OK
{
  "success": true,
  "token": "<jwt>",
  "user": { "_id": "...", "name": "Asha", "email": "user@example.com", "role": "user", "phone": "..." }
}
```

Example: cancel a booking

```
PUT /api/bookings/:id/cancel

200 OK
{ "success": true, "message": "Booking cancelled. Refund initiated.", "booking": { ... } }

400 Bad Request (less than 2 hours before the show)
{ "success": false, "message": "Bookings can only be cancelled up to 2 hours before the show" }
```

Booking flow. `POST /bookings` holds the seats for 10 minutes; `POST /bookings/:id/pay` confirms them. If payment is not made in time the booking becomes `expired`. The demo gateway rejects a card number ending in `0000`, requires a 16-digit card number and a `name@bank` UPI id, and always accepts net banking and wallet.

8. Authentication

- **Passwords** are hashed with bcrypt (10 rounds) before saving and are excluded from queries by default.
- **Tokens.** On login or registration the server signs a JWT containing the account id and role (`user` or `admin`), valid for 7 days by default (`JWT_EXPIRES_IN`).
- **Cookie.** The token is sent in an `httpOnly` cookie named `token` (`secure` and `sameSite=none` in production, `sameSite=lax` in development), so scripts in the page cannot read it. A `Authorization: Bearer <token>` header is also accepted.
- **Authorization.** `protect` verifies the token and loads the account (rejecting deleted or deactivated users); `adminOnly` and `userOnly` enforce roles. Admins have their own collection and login endpoint.
- **Frontend.** `AuthContext` loads the account from `/auth/me`, and `RequireAuth` redirects visitors away from pages that need a login or a specific role.
- **Hardening.** helmet headers, CORS limited to `CLIENT_URL`, and a limit of 100 login/register attempts per IP every 15 minutes.

9. User Interface

Screens are listed with their routes. Screenshots or GIFs can be dropped in under each row once the app is running.

Screen	Route	Access	What it shows
Home	`/`	Public	Featured and current movies
Movies	`/movies`	Public	Search, filters and sorting with movie cards
Movie detail	`/movies/:id`	Public	Cast, rating, reviews, showtimes by city and date
Seat selection	`/shows/:id/seats`	Public	Seat map with live availability, up to 10 seats
Checkout	`/checkout/:id`	User	Hold countdown and payment method form
Booking detail	`/bookings/:id`	Login	E-ticket with booking ID, printable
My bookings	`/my-bookings`	User	History and cancellation

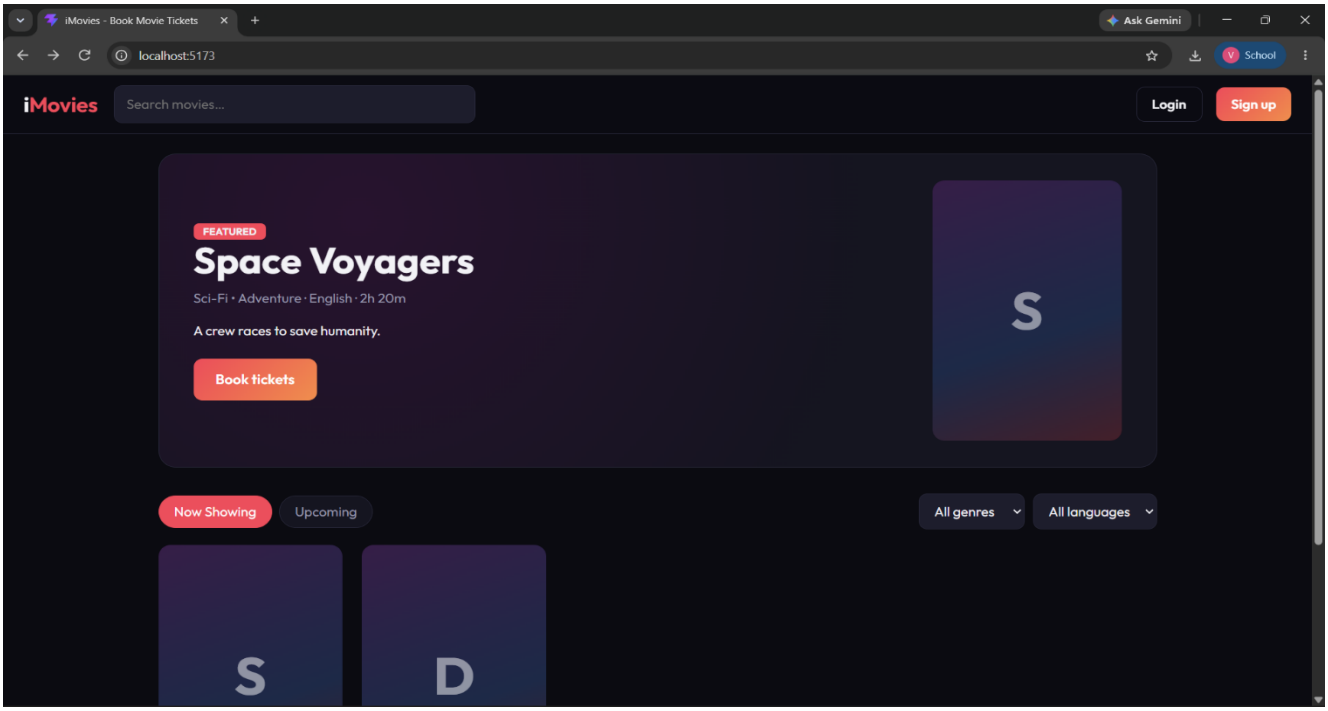
Screen	Route	Access	What it shows
Favourites	/favourites	User	Favourites and watchlist
Profile	/profile	User	Edit details, change password
Login / Register	/login, /register	Public	Customer authentication
Admin login	/admin/login	Public	Admin authentication
Admin panel	/admin	Admin	Dashboard, movies, theatres and screens, shows, users, bookings

10. Testing

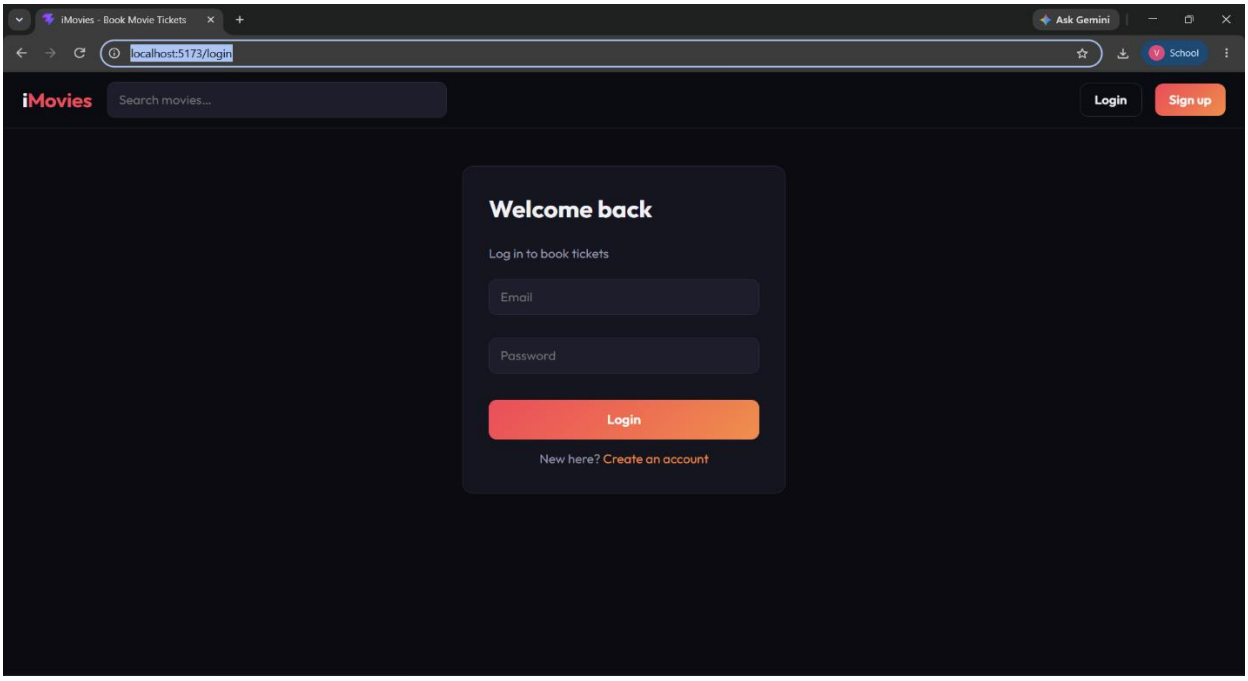
The project has no automated test suite yet; it was checked by hand against the main flows. Tools and checks available today:

- **Manual end-to-end testing** of register, login, browse, seat hold, payment, ticket, cancel, review and favourites, plus every admin screen.
- **Seeded demo data** (npm run seed) gives a repeatable starting state.
- **Payment test cases:** a card ending 0000 is declined; invalid card numbers or UPI ids are rejected; net banking and wallet succeed.
- **API checks** with Postman or curl against /api/health and the endpoints in section 7.
- **ESLint** (npm run lint in client/) for static checks.

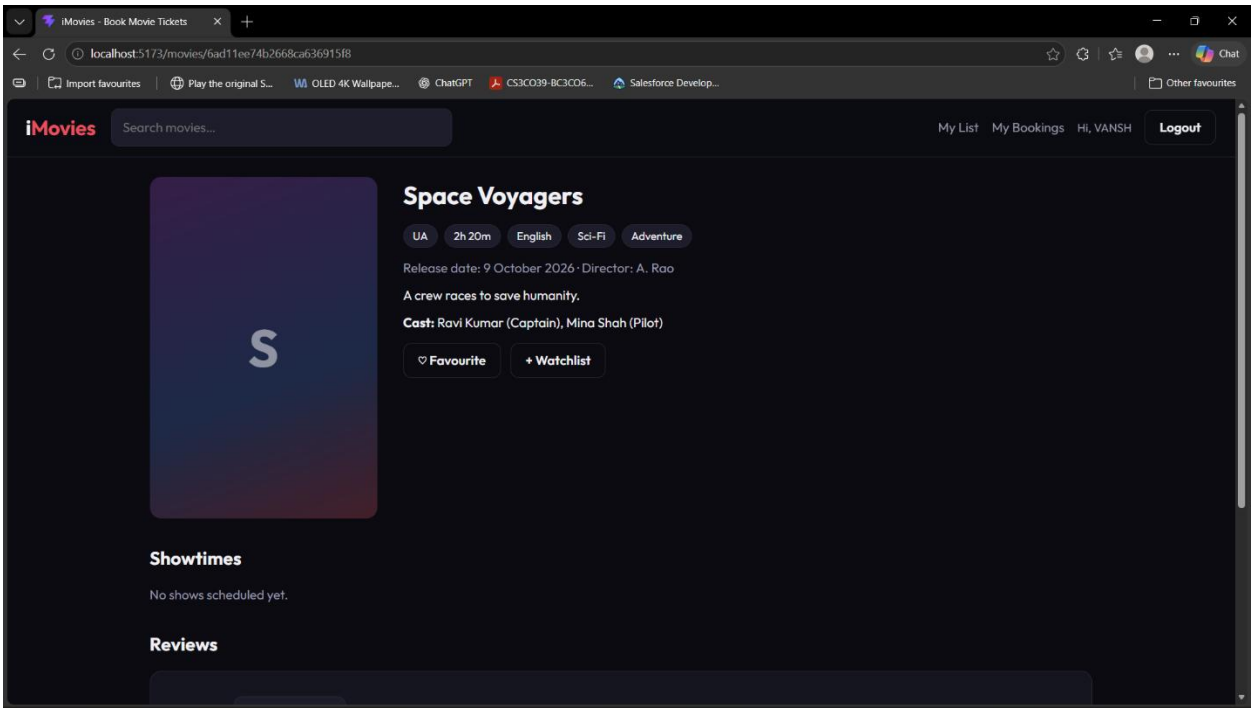
11. Screenshots or Demo



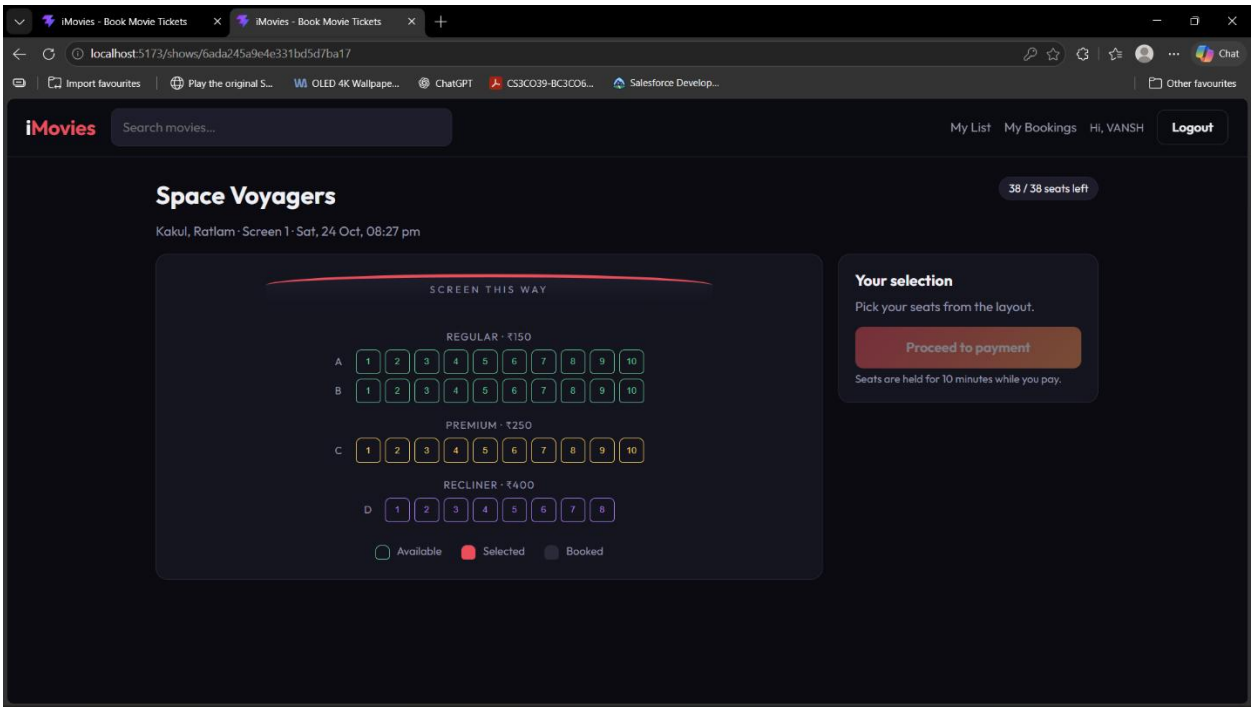
Home Page



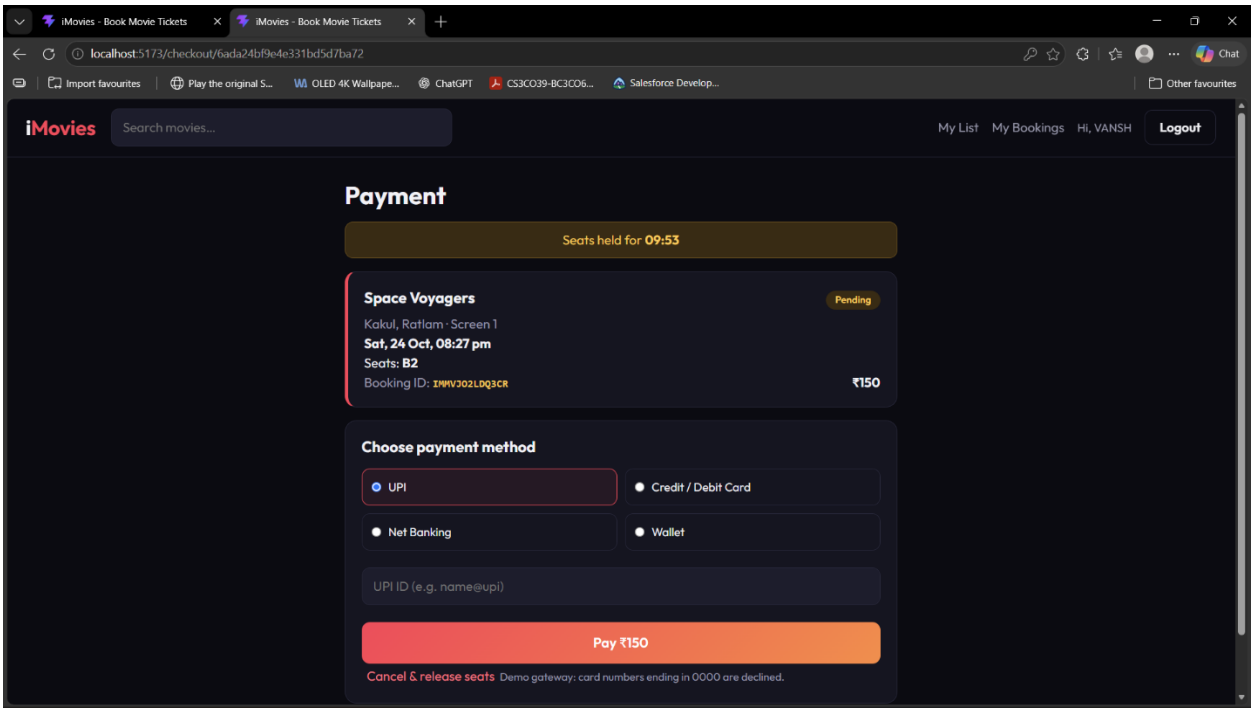
Login Page



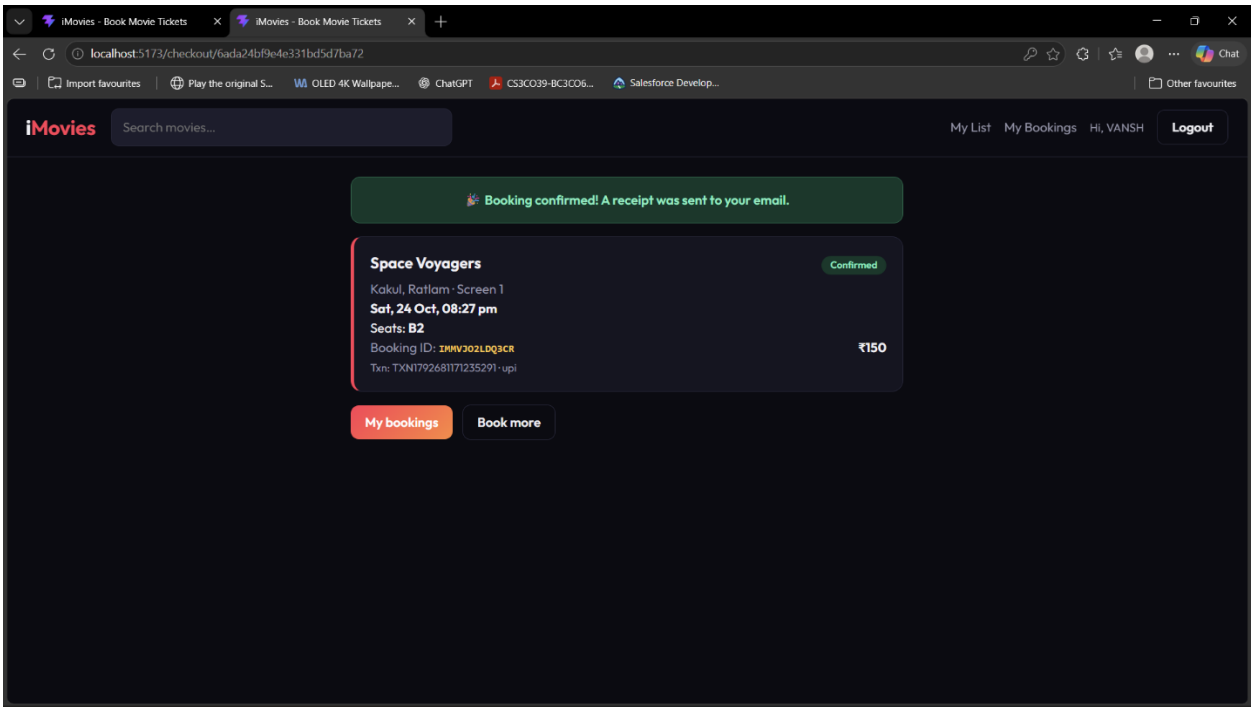
Movie Detail Page



Seat Selection Page



Checkout Page



Confirmation Page

12. Known Issues

- Payments are simulated: no real gateway is connected and refunds are only marked as initiated.
- Emails are printed to the server console unless the `SMTP_*` variables are set.
- The seeded shows are scheduled one day ahead, so "No upcoming shows" appears later until an admin schedules new ones.
- Port 5000 may already be in use (for example macOS AirPlay); change `PORT` in `server/.env` and the proxy target in `client/vite.config.js`.
- The seeded admin password is public in the README and must be changed before any real deployment.
- No automated tests exist.

13. Future Enhancements

- Real payment gateway (for example Razorpay or Stripe) with actual refunds
- Automated unit and API tests, and CI
- SMS and email ticket delivery with a QR code
- Poster and image upload by admins instead of URLs
- Food and beverage add-ons, coupons and offers
- Movie recommendations based on favourites and past bookings
- Multi-language interface and a mobile app